

Mini Hybrid V2.2 · 2026-09-29 · generated from configs/V2.2_sector_40x1.20mm_valve3.8s.json

Current design

injector 40 x Ø1.20 mm THRU, 12.7 mm plate (L/D 10.58, HEM), Cd 0.70 assumed
sized on chamber pressure, 25 bar target (does not invert the regression law)
grain centre port Ø14.14 + 4 sector ports, OD 85.85, L 177.8
webs: spoke/ring 17.53 mm, wall 8.76 mm; port area 1241 mm² (1.70 x throat)
shutdown run valve closes at 3.8 s; liquid runs out at 4.05 s
tank 7.0 L, 4.48 kg N2O at 23.9 °C, self-pressurised
nozzle throat Ø30.48 mm, ε 4.34

What it does, under the two regression fits that are most likely N2O data

fit	Pc peak	Pc mean	F mean	impulse	Isp	O/F	web used
SP7 a	25.5 b	23.6 b	2121 N	8058 Ns	208s	6.54	7.08 of 8.76 mm
SP7 b	24.2 b	23.0 b	2067 N	7854 Ns	207s	7.75	6.16 of 8.76 mm

The decisions, and why

- Size on chamber pressure, not O/F. The O/F objectives invert the regression law with exponent $1/(1-n) \approx 1.4\text{-}3.1$; across the candidate fits a burn-average O/F target asks for 14 to 10084 holes. Sizing on 25 bar gives 39-44 holes over the same 9x spread in 'a'. The plate is now the settled part of the design.
- Keep Ø1.20 mm holes in the 12.7 mm plate. L/D 10.6 puts the orifices in the HEM regime; varying the count rather than the diameter keeps the flow model valid.
- Sector-port grain. At the same port area (so the same starting oxidiser flux) it gives an 8.76 mm wall web, against 7.00 mm for the five round ports on their best circle and 5.42 mm for a centre + 4 layout.
- Close the run valve before the liquid runs out. The vapour tail delivers ~13% of the impulse but burns a fifth to a quarter of the web; cutting it keeps both likely-N2O fits inside the wall web with ≥ 1.7 mm to spare.

Regression coefficients are literature values. Everything that depends on them — O/F, web use, Isp — is a range until the first fire is weighed.

	revision 1	revision 2 (now)
sizing objective	oxidiser flow 1.0 kg/s	chamber pressure 25 bar
plate	41 x Ø1.20, 12.7 mm, HEM	40 x Ø1.20, 12.7 mm, HEM
grain ports	5 x Ø17.78 round	Ø14.14 centre + 4 sectors
wall web	10.31 nominal (7.00 real)	8.76 exact
burning surface	497 cm ²	625 cm ² (+26%)
shutdown	blow down to empty	valve closes at 3.8 s
regression law	HRAP a 0.0304, n 0.681	range of 6 fits, 2 likely
c* / table	HRAP Paraffin.mat	same

Why the regression coefficients were demoted

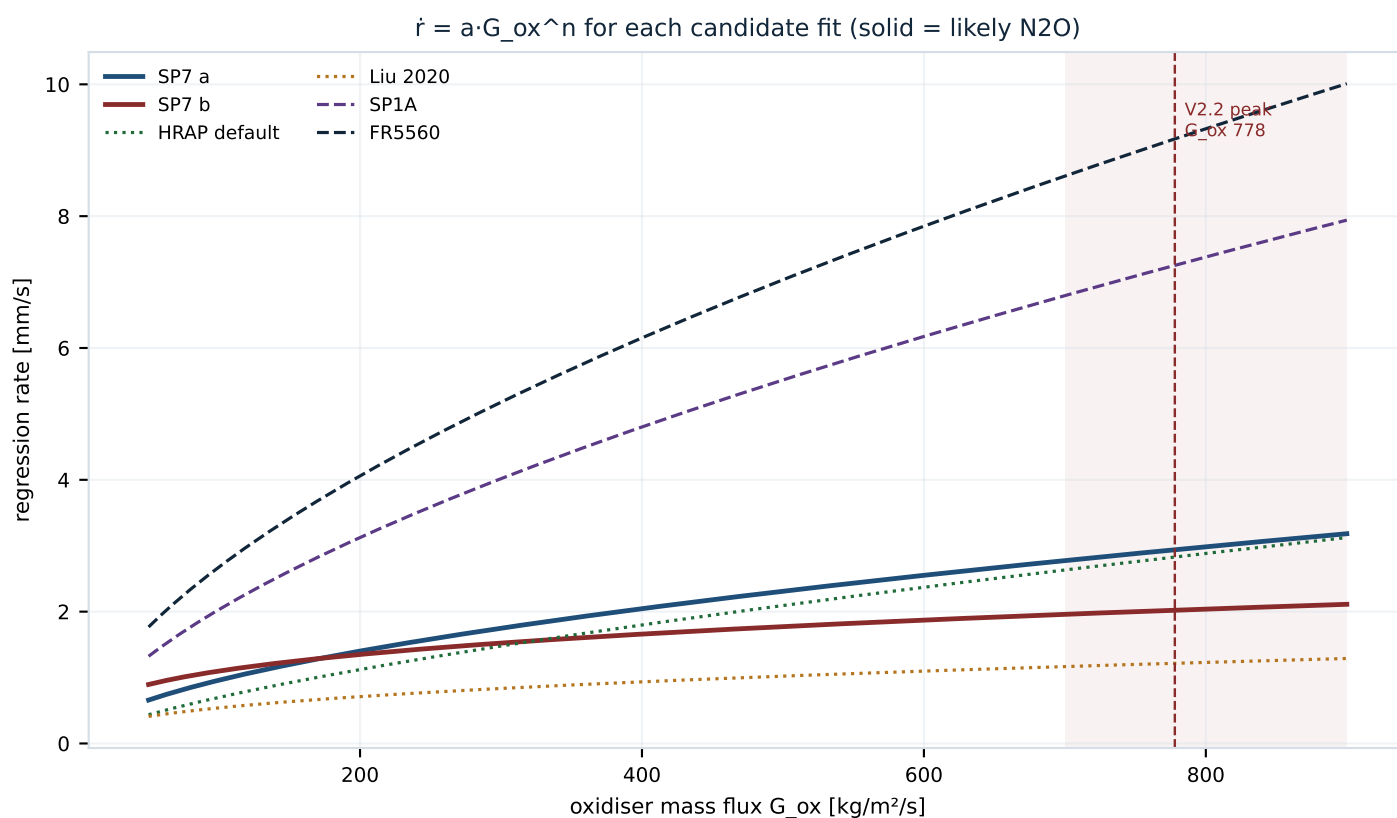
HRAP's 'paraffin' pair (a 0.0304, n 0.681) matches the HTPB (Thiokol) row of Karp & Jens Table 6.2 to three significant figures, and HRAP ships the same pair for ABS, asphalt and HTPB. The PDF of that table drops subscripted formulas, so its oxidiser column is blank; the N2O rows were identified by converting Doran et al. (AIAA 2007-5352) N2O coefficients for HDPE, PMMA and HTPB to SI — all three match the table exactly, and the two SP7 paraffin rows sit in the same block. SP1A, which looked like the obvious N2O row, is probably GOX: Doran declines to report SP1A coefficients at all.

Why the web became the question

With the plate fixed on chamber pressure, the uncertainty moved to the grain. The tool's multi-port web was an equal-area figure (10.31 mm); the five round ports on their best circle actually leave 7.00 mm to the case, and both SP7 fits regress 9.5-9.6 mm over a full blowdown. The round-port grain burns through under the most likely fuel data. Revision 2 answers that with the port layout and the shutdown, not with the injector.

Tool changes behind these numbers

- Sector grain with exact area/perimeter at every regression depth; balanced-web generator; grain drawing sheet and DXF.
- Commanded valve closure; web-used reporting; warnings for burn-through, thin web margin, nominal multi-port web and ports smaller than the throat.
- Transient chamber model now timestep-converged; injector ΔP measured against the real feed pressure; flux-range warning on the regression law.
- Fuel presets carry oxidiser, flux range and source; placeholder coefficients are flagged when used with an O/F objective.



fit	a	n	oxidiser	$\dot{r}@150$	@300	@500	@700	mm/s
SP7 a	0.0781	0.545	N2O (very likely)	1.20	1.75	2.31	2.77	
SP7 b	0.2800	0.297	N2O (very likely)	1.24	1.52	1.77	1.96	
HRAP default	0.0304	0.681	none -- HTPB	0.92	1.48	2.09	2.63	
Liu 2020	0.0876	0.395	N2O, G 91-242	0.63	0.84	1.02	1.17	
SP1A	0.1170	0.620	probably G0X	2.61	4.02	5.52	6.79	
FR5560	0.1690	0.600	unconfirmed	3.42	5.18	7.04	8.61	

Reading it

- Compare fits at the same flux, never by 'a'. The units of a depend on n, so a 0.28 with n 0.30 and a 0.078 with n 0.55 are not comparable numbers. The two SP7 fits agree within 15% at 300 kg/m²/s and part to 42% at 700 — they differ mainly in n, which only a second fire separates.
- Against HRAP's default at 300 kg/m²/s: SP7 a 1.18x, SP7 b 1.03x, SP1A 2.7x, FR5560 3.5x. The likely-N2O data sit close to the placeholder; the fast fits, which are probably not N2O, are the ones far from it.
- Liu 2020 is a metallised paraffin fitted at 91-242 kg/m²/s; at V2.2's 300-780 it is extrapolated and reads low. Treat it as a lower bound.
- V2.2 starts at 778 kg/m²/s, above the ~700 upper end of the published fits. The early burn is extrapolation for every fit. G_{ox} is set by port area, which the sector grain kept equal to preserve this number rather than make it worse.

Objective robustness — the reason for sizing on Pc

fit	Pc 25 bar	O/F 7.5 (burn avg)
SP7 a	39 holes	64
SP7 b	41 holes	38
HRAP default	39 holes	40
Liu 2020	44 holes	14
SP1A	41 holes	6124 degenerate
FR5560	39 holes	10084 degenerate

Across a 9x spread in 'a' the chamber-pressure objective moves the hole count from 39 to 44. The O/F objective has to invert the regression law, so the same spread moves it by orders of magnitude, and at the fast fits it drives chamber pressure onto tank pressure (degenerate). Sizing on Pc and reading O/F off afterwards is the well-conditioned direction.

Chamber-pressure target sweep (sector grain, valve 0.25 s before liquid-out)

Pc	holes a/b/H	valve	Pc pk	F mean	impulse	Isp	min ΔP	web a	web b	G pk
20	31/32/31	4.98	19.8	1588	7904	198	149%	89%	83%	603
25	39/41/39	3.90	24.8	2063	8043	207	100%	82%	71%	758
30	47/51/47	3.20	29.8	2526	8082	212	68%	76%	63%	914
35	55/60/55	2.70	34.7	2980	8043	216	45%	71%	57%	1069

Higher chamber pressure needs more holes and buys Isp through the nozzle, but shortens the liquid phase, raises oxidiser flux further past the fitted range and eats injector pressure drop. 25 bar keeps ΔP near 100% of Pc all burn with the web inside margin for both N2O fits. 30 bar is a candidate once the regression rate is measured; 35 bar is not, on this tank and case.

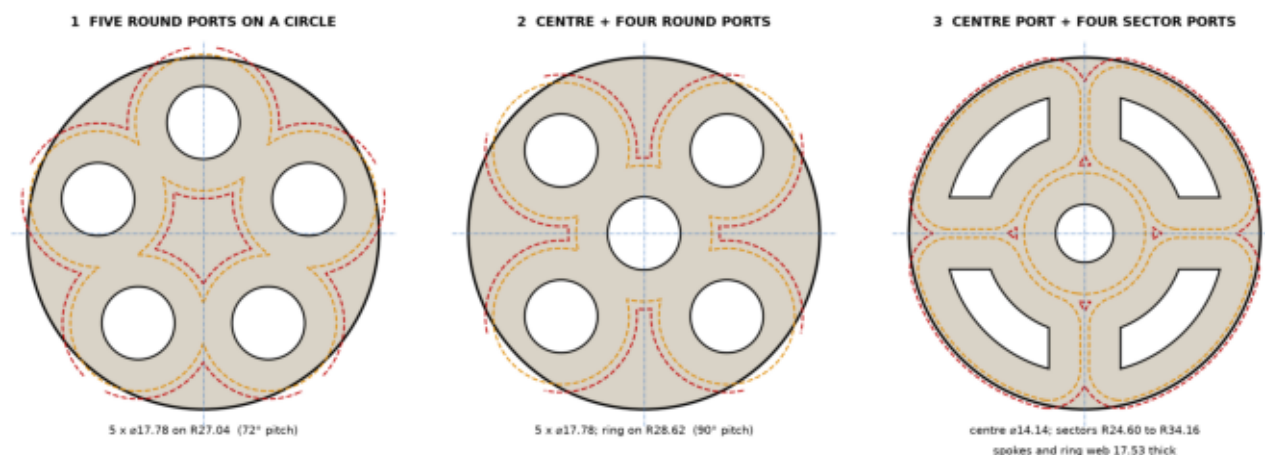
Discharge coefficient — the dominant plate uncertainty now

Cd	holes for 25 bar
0.60	46
0.70	39
0.80	34

Cd moves the hole count more than the regression fits do. A water cold flow of the drilled plate measures it in an afternoon; drill fewer holes first and open up after the cold flow if needed.

V2.2 FUEL GRAIN - PORT LAYOUT OPTIONS (cross-section, mm)

ø85.85 OD x 177.8 long paraffin. All three have the same total port area (1241 mm²), so the same initial oxidiser flux and fuel load.



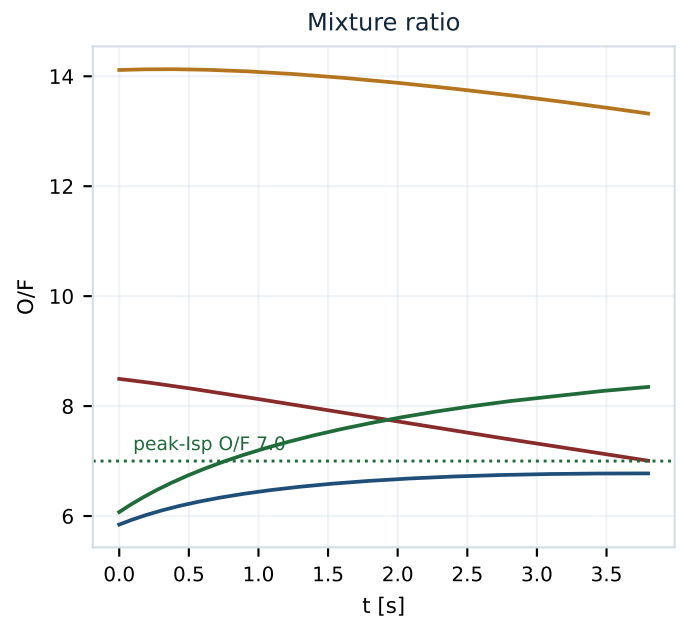
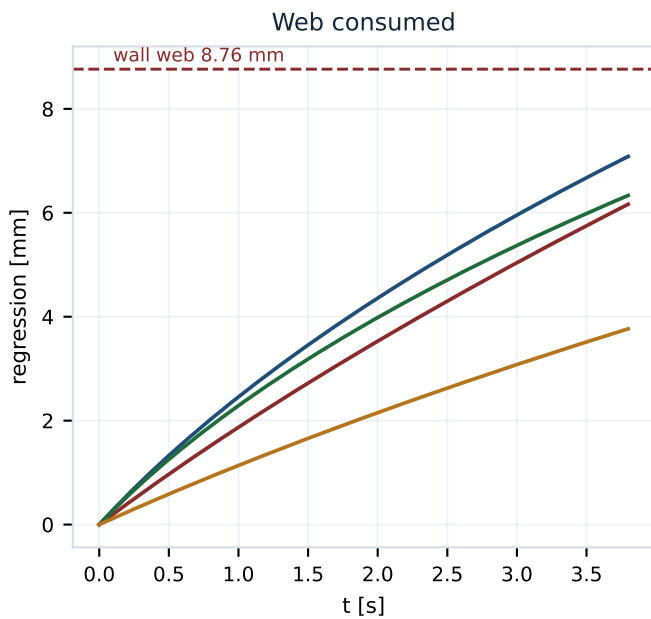
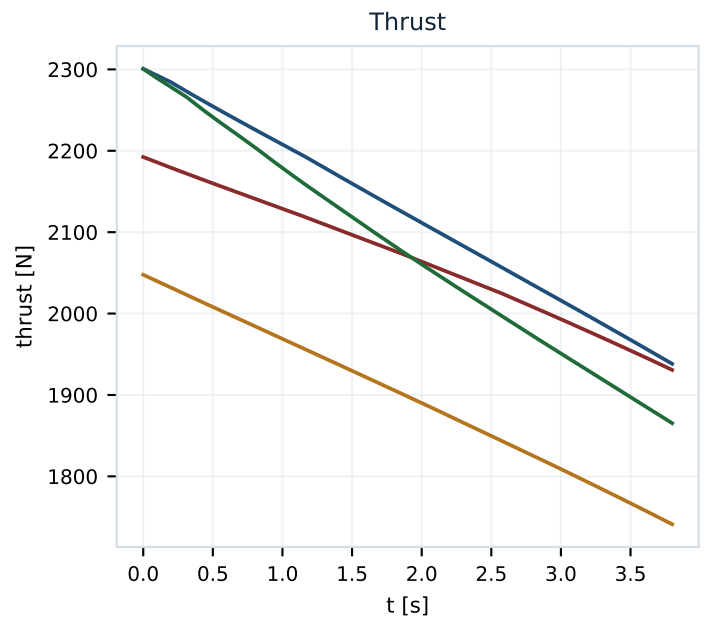
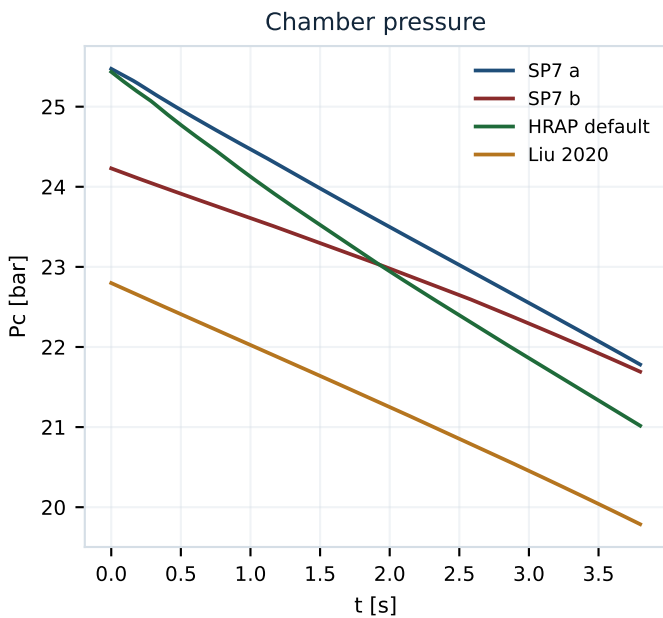
Cross-sections as compared earlier (dashed: SP7 a regression at liquid-out and at end of a full blowdown). Table: 40-hole plate, valve closing at 3.8 s.

layout	wall	area	perim	A/At	G pk	SP7 a web	SP7 b web	O/F a
5 round on a circle	7.00	1241	279	1.70	778	7.36	OUT	6.30 ok 7.37
centre + 4 round	5.42	1241	279	1.70	778	7.36	OUT	6.30 OUT 7.37
sector, same area	8.76	1241	351	1.70	778	7.08	ok	6.16 ok 6.54
sector, same surface	10.56	689	279	0.94	1401	8.14	ok	6.78 ok 6.46

- The sector layout is the only one that keeps both N2O fits off the case at the same port area. It does it by using the diameter better: round ports leave fuel between them that does not protect the wall.
- Its cost is 26% more burning surface: more fuel flow and lower O/F (6.5/7.8 for SP7 a/b, against 7.4/9.0 for the round grain). The CEA table peaks at O/F 7.0 and is flat near the top, so the Isp cost is small.
- Shrinking the sector ports to match the round grain's surface gives the most web but halves the port area — below the throat area — and nearly doubles oxidiser flux. The grain would choke the flow. Rejected.
- Round-port rows are simulated with nominal growth and judged against the real wall web of their layout.

fit	full blowdown		valve at 3.8 s		3.5 kg loaded	
	web mm	Ns	web mm	Ns	web mm	Ns
SP7 a	THRU	9712	7.08	8058	7.87	7519
SP7 b	THRU	9574	6.16	7854	7.91	7320
HRAP default	7.94	9632	6.33	7890	6.75	7428
Liu 2020	5.36	8854	3.77	7206	4.50	6791

- A full blowdown burns through the 8.76 mm wall under both SP7 fits.
- Closing the valve at 3.8 s keeps every fit here inside the web and keeps 81-83% of the full-blowdown impulse; the vapour tail it removes is low-pressure and low-flux, which is also where chuffing and O/F drift live.
- Loading 3.5 kg and letting it blow down is worse on both counts: it still ends with a vapour tail, so it uses more web and delivers less impulse than the cutoff.
- The valve closes on ~0.2 kg of remaining liquid. Liquid runs out at 4.05 s in the model; that moves with fill temperature and real Cd, so confirm it from the tank-pressure knee in cold flow, and vent the ~1 kg left in the tank remotely after the fire.



Peak chamber pressure spans only 22.8-25.5 bar across four fits a factor of ~ 3 apart in regression rate — it is set mostly by the injector and the tank, which is the point of sizing on P_c . What the fit moves is O/F (6.5-7.8 for the N2O fits, 13.9 for Liu) and how much web is used. Web use is the number to watch on the first fire.

Even the direction of O/F drift depends on n : it rises through the burn for the high- n fits (SP7 a, HRAP) and falls for the low- n ones (SP7 b, Liu). The likely-N2O fits stay within about 1.5 of the peak-Isp O/F throughout, which is on the flat top of the Isp curve.

1. The nozzle is already matched — chamber pressure is the lever

ϵ 4.34 is perfectly expanded at 24.9 bar and the motor averages 23.6 bar; the ideal ϵ at that pressure is 4.19, worth +0.0% in C_f . Revision 1 ran at ~20 bar and was over-expanded; sizing to 25 bar closed that gap. Re-cutting the nozzle for the current design buys nothing.

The remaining nozzle-side gain is a higher chamber pressure: the P_c sweep gives 207 s at 25 bar and 216 s at 35 bar (+4.1%), more with ϵ re-matched. Its price is injector ΔP margin (100% $\rightarrow$ 45% of P_c), a shorter liquid phase and flux further past the fitted range, so it is a step for after the regression rate is measured, not before.

2. Measure the regression rate, then re-optimize

One fire with the grain weighed before and after fixes 'a' for an assumed n ; a second at a different flow separates a from n . Every O/F, I_{sp} and web number in this report collapses from a range to a value, and the valve time can move toward liquid-out (or the load up) with a known margin.

3. Grain options, if the case can change

- Larger grain OD: every millimetre of radius is wall web, and more port area lowers the 778 kg/m²/s starting flux into the fitted range.
- Longer grain: more burning surface at the same flux raises fuel flow toward the I_{sp} -peak O/F without shrinking ports — and paraffin's regression does not depend on length ($m = 0$).
- Pre- and post-combustion chambers: paraffin burns by droplet entrainment, and mixing volume raises c^* efficiency, which is assumed 1.0 here and will be lower.
- Fillets on the sector corners (R2 costs ~1% port area) and a phenolic liner for insulation margin against the case.

4. Feed system

- Regulated or supercharged feed would hold chamber pressure and ΔP flat through the burn (revision 1 found 65 bar supercharge the best single result), but it is a different pressure system and does nothing for flux.
- Fill-temperature control mainly buys repeatability: the liquid-out time, and so the valve timing, moves with it.

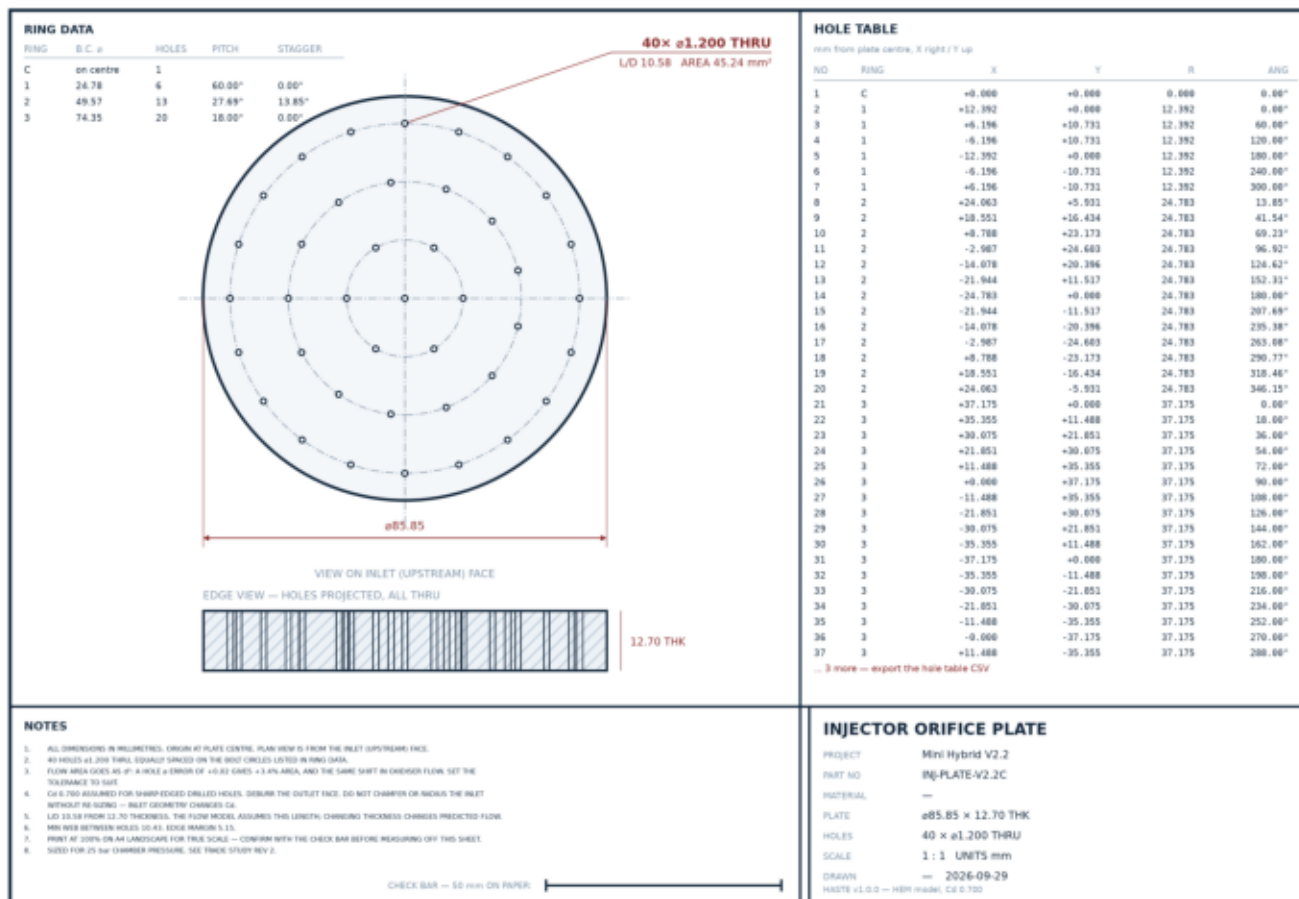
5. Injector

The showerhead plate is sized and drawn. Beyond C_d , the open question is atomisation and mixing; impinging or swirl elements would help c^* efficiency more than any change to the hole count.

1. water cold flow of the drilled plate -> Cd; re-size (36-48 holes span)
2. N2O cold flow, tank pressure logged -> liquid-out time; confirms HEM
3. set the valve timer from step 2 -> ~0.25 s before the pressure knee
4. first fire, grain weighed before/after -> a (n assumed); web margin check
5. second fire at another flow -> separates a from n
6. re-run this study with measured values -> nozzle ϵ , valve time, ox load

Known model limits at this operating point

- Oxidiser flux starts above every fitted range; early-burn regression is extrapolation.
- c^* efficiency 1.0 and nozzle efficiency 0.9 are assumptions; delivered I_{sp} will be lower.
- No combustion-instability model: the 20% ΔP rule is guidance, not a prediction.
- Valve closure is instantaneous and post-shutdown smoulder is neglected.
- Blowdown is adiabatic and in equilibrium; real tanks decay a little more slowly.
- Sector corners are sharp in the model; fillets remove ~1% of port area.
- SP1A and FR5560 burn through in about 2 s whatever the valve does. Both are probably not N2O data, but the first fire is what rules them out.



40 x Ø1.20 mm on rings of 6/13/20, 12.7 mm plate. Full-size sheet, DXF and hole table: Trade Study/V2.2_injector_plate_40x1.20mm.*

